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MySQL Data Types

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MySQL Data Types (Version 8.4)

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The data type of a column defines what value the column can hold: integer, character, money, date and time, binary, and so on.

Each column in a database table is required to have a name and a data type.

An SQL developer must decide what type of data that will be stored inside each column when creating a table. The data type is a guideline for SQL to understand what type of data is expected inside of each column, and it also identifies how SQL will interact with the stored data.

In MySQL there are three main data types: String, Numeric, and Date and Time.

String Data Types

Data type	Description

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VARCHAR(size)	A variable-length string (can contain letters, numbers, and special characters). *size* specifies the maximum column length in characters, from 0 to 65535
BINARY(size)	Similar to CHAR(), but stores binary byte strings. *size* specifies the column length in bytes. Default *size* is 1
VARBINARY(size)	Similar to VARCHAR(), but stores binary byte strings. *size* specifies the maximum column length in bytes.
BLOB(size)	A BLOB column with a maximum length of 65535 bytes
TINYBLOB	A BLOB column with a maximum length of 255 bytes
MEDIUMBLOB	A BLOB column with a maximum length of 16777215 bytes
LONGBLOB	A BLOB column with a maximum length of 4294967295 or 4GB bytes
TEXT(size)	Holds a string with a maximum length of 65535 bytes
TINYTEXT	A TEXT column with a maximum length of 255 characters
MEDIUMTEXT	A TEXT column with a maximum length of 16777215 characters
LONGTEXT	A TEXT column with a maximum length of 4294967295 or 4GB bytes
ENUM(val1, val2, val3, ...)	A string object that can have only one value, chosen from the list of possible values (val1, val2, val3,..). You can list up to 65535 values. If a value is inserted that is not in the list, a blank value will be inserted
SET(val1, val2, val3, ...)	A string object that can have zero or more values, chosen from the list of possible values (val1, val2, val3,..). You can list up to 64 values

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Numeric Data Types

Data type	Description
BIT(*size*)	A bit-value type. *size* indicates the number of bits per value, from 1 to 64. The default value for *size* is 1.
TINYINT(*size*)	A very small integer. Signed range is from -128 to 127. Unsigned range is from 0 to 255. *size* specifies the minimum display width (max display width is 255)
BOOL	A value of zero is considered as false, nonzero values are considered as true.
BOOLEAN	Equal to BOOL
SMALLINT(*size*)	A small integer. Signed range is from -32768 to 32767. Unsigned range is from 0 to 65535. *size* specifies the minimum display width (max display width is 255)
MEDIUMINT(*size*)	A medium-sized integer. Signed range is from -8388608 to 8388607. Unsigned range is from 0 to 16777215. *size* specifies the minimum display width (max display width is 255)
INT(*size*)	A normal-size integer. Signed range is from -2147483648 to 2147483647. Unsigned range is from 0 to 4294967295. *size*

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BIGINT(*size*)	A large integer. Signed range is from -9223372036854775808 to 9223372036854775807. Unsigned range is from 0 to 18446744073709551615. *size* specifies the minimum display width (max display width is 255)
DECIMAL(*size*, *d*)	An exact fixed-point number. *size* indicates the total number of digits (max 65). *d* indicates the number of digits after the decimal point (max 30). The default value for *size* is 10. The default value for *d* is 0.
DEC(*size*, *d*)	Synonym for DECIMAL(*size*, *d*)
FLOAT(*size*, *d*)	A small floating-point number. FLOAT(*size*, *d*) is deprecated in MySQL 8.0.17, and it will be removed in future MySQL versions
FLOAT(*p*)	A floating-point number. The *p* value determines whether to use FLOAT or DOUBLE for the resulting data type. If *p* is from 0 to 24, the data type becomes FLOAT. If *p* is from 25 to 53, the data type becomes DOUBLE
DOUBLE(*size*, *d*)	A normal-size floating-point number. Permissible values are -1.7976931348623157E+308 to -2.2250738585072014E-308, 0, and 2.2250738585072014E-308 to 1.7976931348623157E+308. The total number of digits is specified in *size*. The number of digits after the decimal point is specified in the *d* parameter
DOUBLE PRECISION(*size*, *d*)	Synonym for DOUBLE(*size*, *d*)

Date and Time Data Types

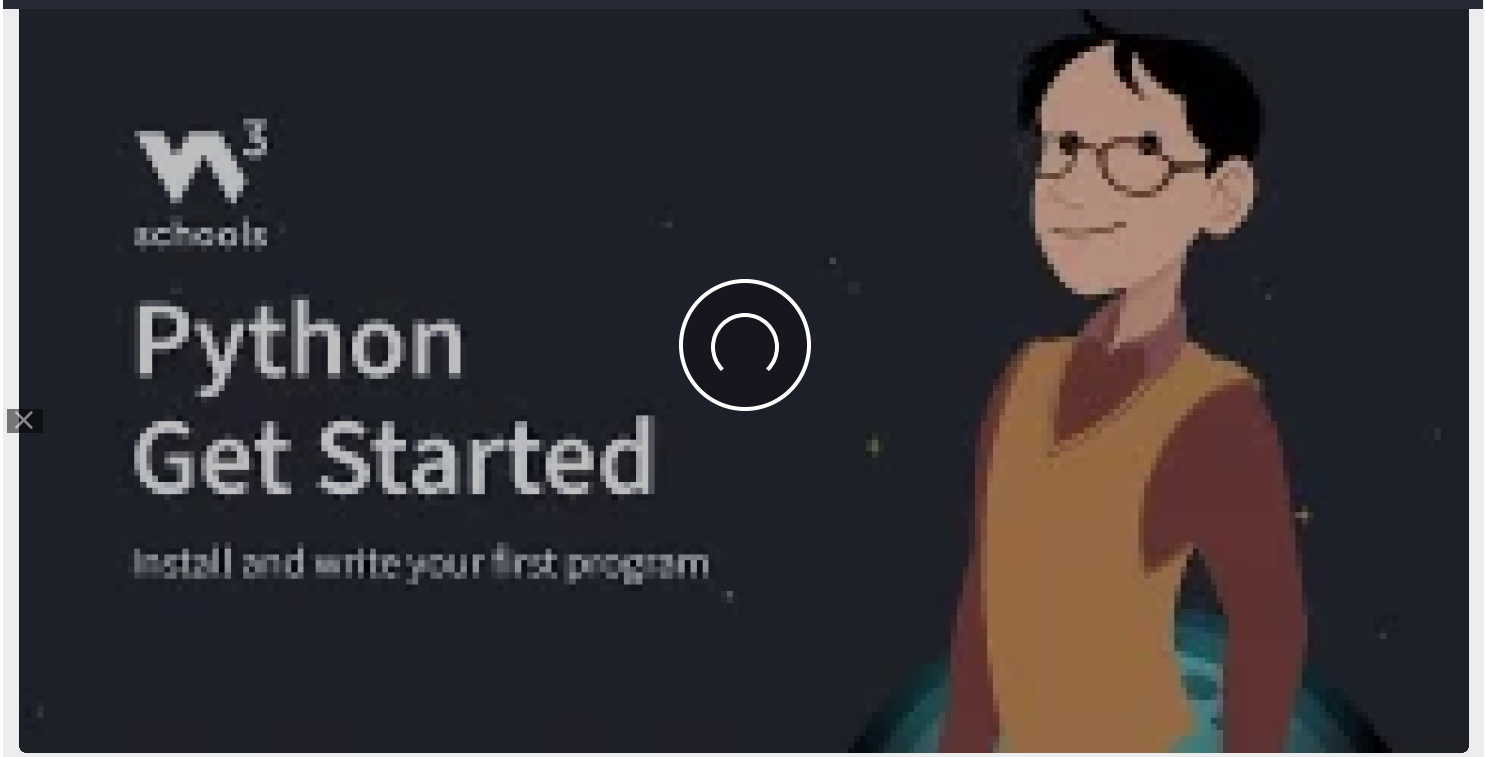
Data type	Description
DATE	A date. Format: YYYY-MM-DD. The supported range is from '1000-01-01' to '9999-12-31'
DATETIME(*fsp*)	A date and time combination. Format: YYYY-MM-DD hh:mm:ss. The supported range is from '1000-01-01 00:00:00' to '9999-12-

TIMESTAMP(*fsp*)	A timestamp. Format: YYYY-MM-DD hh:mm:ss. TIMESTAMP values are stored as the number of seconds since the Unix epoch ('1970-01-01 00:00:00' UTC). The supported range is from '1970-01-01 00:00:01' UTC to '2038-01-09 03:14:07' UTC. Automatic initialization and updating to the current date and time can be specified using DEFAULT CURRENT_TIMESTAMP and ON UPDATE CURRENT_TIMESTAMP in the column definition
TIME(*fsp*)	A time. Format: hh:mm:ss. The supported range is from '-838:59:59' to '838:59:59'
YEAR	A year in a four-digit format (YYYY). Values allowed in four-digit format: 1901 to 2155, and 0000.

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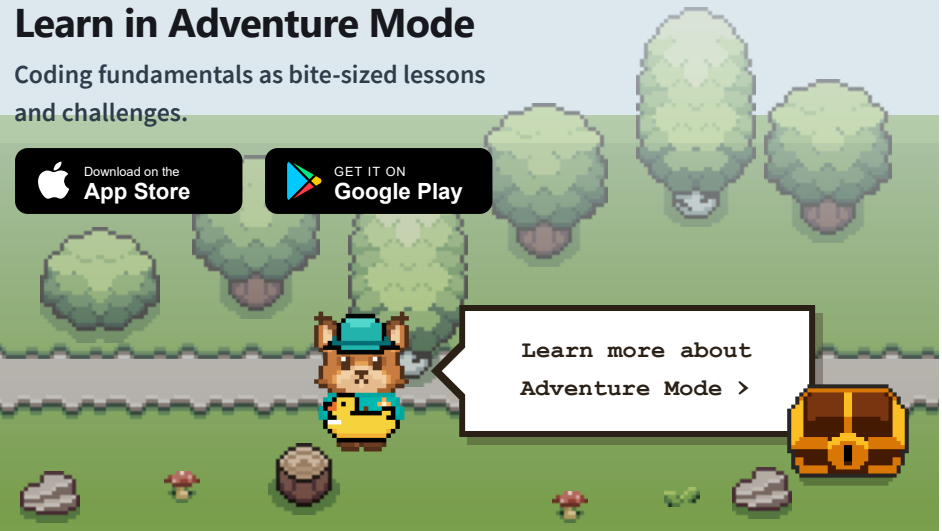
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